

CCSDS Convolutional Encoder ($r=1/2$, $K=7$)

What it is and exactly where it lives in a TM/AOS TX chain

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1 What the CCSDS $r=1/2$, $K=7$ encoder is

Type. Non-systematic, “transparent” convolutional code, rate $1/2$, constraint length $K = 7$ (memory 6), decoded with Viterbi.

Generators. $G_1 = 171_8$, $G_2 = 133_8$, with an inversion on the G_2 output for the *base* $r = 1/2$ mode (punctured modes omit the inverter).

Output bit multiplexing. $C_1(1)$, $C_2(1)$, $C_1(2)$, $C_2(2)$, $\dots$

Puncturing (optional). The standard defines punctured rates ($2/3$, $3/4$, $5/6$, $7/8$) derived from the base code. These modes do not use the G_2 inverter.

2 Where it sits in the TX chain (order matters)

For the classic concatenated scheme (RS outer, convolutional inner), the encoder belongs *after* RS and *after* ASM insertion, just before modulation:

TM Transfer Frame $\xrightarrow{\text{(optional) Randomize}}$ RS(255,223), interleave depth I $\xrightarrow{\text{Insert ASM (0x1ACFFC1D)}}$ Conv. $r=1/2$, $K=7$

Why this order

- **Concatenation rule.** RS is the *outer* code and the convolutional code is the *inner* code; i.e., RS output feeds the convolutional encoder.
- **ASM placement.** The Attached Sync Marker (ASM) precedes the transmitted codeblock and forms a CADU = ASM + RS codeblock.
- **Randomizer timing.** Randomize the RS codeblock only (TM Transfer Frame + RS parity), *not* the ASM. In practice the pseudo-random sequence is applied to each transmitted RS codeblock; the ASM is excluded.

3 So in practice your TX should

1. Build the TM frame.
2. Randomize the TM frame bytes (if using PRN).
3. RS(255,223) encode and interleave (depth I).
4. Prepend the 32-bit ASM (0x1ACFFC1D).

5. Feed ASM + RS codeblock to the $r = 1/2$, $K = 7$ convolutional encoder.
6. Serialize/map bits to symbols, then pulse-shape and modulate.

4 Practical details you'll care about

Bit ordering. Follow $C_1(1), C_2(1), C_1(2), C_2(2), \dots$ as you pack toward bytes / symbol mapping.

G_2 inversion. Ensure the second output stream (from G_2) is inverted in the base $r = 1/2$ implementation. Punctured modes omit this inverter.

Sync acquisition. Because the ASM is inserted before the inner code, it is *convolutionally encoded*. Many receivers (a) Viterbi-decode first and then correlate on the raw ASM, or (b) correlate on a pre-encoded ASM to aid Viterbi sync. In all cases, the ASM is the cadence anchor for RS codeblock sync.

Randomizer init & scope. Initialize the PRN at the start of each RS codeblock and run across the full RS length; do not randomize the ASM.

Why randomize? The pseudo-randomizer increases transition density, which helps the inner code and downstream modulation maintain spectral and timing properties.

5 Minimal RX counterpart (for completeness)

Demod $\rightarrow$ (optionally) ASM-assisted sync $\rightarrow$ Viterbi $r=1/2 \rightarrow$ Find/strip ASM $\rightarrow$ RS de-interleave & decode $\rightarrow$ D

Standards emphasize that RS decoding is aligned using the ASM cadence, and the PRN derandomizer is synchronized from the ASM boundary.

6 Appendix: quick reference table

Concept / Quantity	Definition / Note
Conv. code	$r = 1/2$, $K = 7$, $G_1 = 171_8$, $G_2 = 133_8$, G_2 output inverted (base mode)
Output order	$C_1(1), C_2(1), C_1(2), C_2(2), \dots$
Punctured modes	$2/3$, $3/4$, $5/6$, $7/8$ derived from base; no G_2 inverter
ASM	$0x1ACFFC1D$ (32 bits), precedes RS block, not randomized
RS outer code	RS(255,223) over GF(256), interleave depth I (e.g., $I = 5$)
Randomizer	PRN applied across RS codeblock (excluding ASM), reinit each block
CADU	ASM + RS codeblock (byte-wise), then inner conv. coding (bit-wise)

References (useful standards/ICDs)

- ECSS-E-ST-50-01C (European *adoption* of CCSDS TM Synchronization & Channel Coding). Contains the $r = 1/2$, $K = 7$ code (171/133 with G_2 inversion), concatenation (RS outer / Conv inner), ASM usage, and randomizer details.

- CCSDS 131.0 (TM Synchronization and Channel Coding), Blue Book.
Base normative reference for the code, ASM, and concatenation.
- NOAA JPSS X-band ICD (Rev. C).
A modern ICD explicitly stating: ASM precedes the randomized RS codeblock; ASM is not randomized; PRN polynomial/initialization; reiterates $r = 1/2$, $K = 7$, 171/133.